



CRUDE DISTILLATION UNIT · INTEGRITY CASE STUDY

From Failure to Control

Root cause analysis · risk register · permit to work · job safety analysis

One unit. One causal chain. Four deliverables that hand over to each other.

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TWELVE MONTHS ON ONE CRUDE DISTILLATION UNIT

Three failures. Nobody was hurt.

\$23.1M

Direct cost

lost margin and repair

28

Days of lost production

across three events

3

High-potential near misses

H₂S, pool fire, firebox fire

0

Injuries

not entirely by design

The zero is not a result. Two of these three could have killed somebody and did not, for reasons that were not under our control. A hot oil release found no ignition source. A tube ruptured when nobody was standing at the heater. That is luck, and luck is not a barrier.

Figures are indicative, from a constructed teaching case — see Basis and Scope in the handout.

Backward look, forward look, authorisation, execution.

Each deliverable produces what the next one consumes. That is the whole session in one line.



→ what we learn becomes tracked risk

→ tracked risk generates work on live plant

→ the permit authorises the task

01

THE BACKWARD LOOK

Why three things failed

Three different mechanisms — aqueous corrosion, mechanical reliability, high-temperature metallurgy. One analysis template.

The same seven questions, asked of every failure.

The repetition is deliberate. By the third failure you will be predicting the structure and listening to the content.

A

Event and consequence

What happened, what it cost, who could have been hurt

B

Evidence preserved

What we secured before anything was cleaned or repaired

C

Physical root

What the metal actually did, named per API RP 571

D

Human root

What people did, or did not do, and why it was reasonable to them

E

Latent / system root

What the organisation permitted or never asked for

F

Barrier analysis

What should have stopped it — absent, degraded, or working

G

Corrective actions

Ranked against the hierarchy of controls, not listed at random

Section E is where the value is. A report that stops at C and D is a maintenance record.

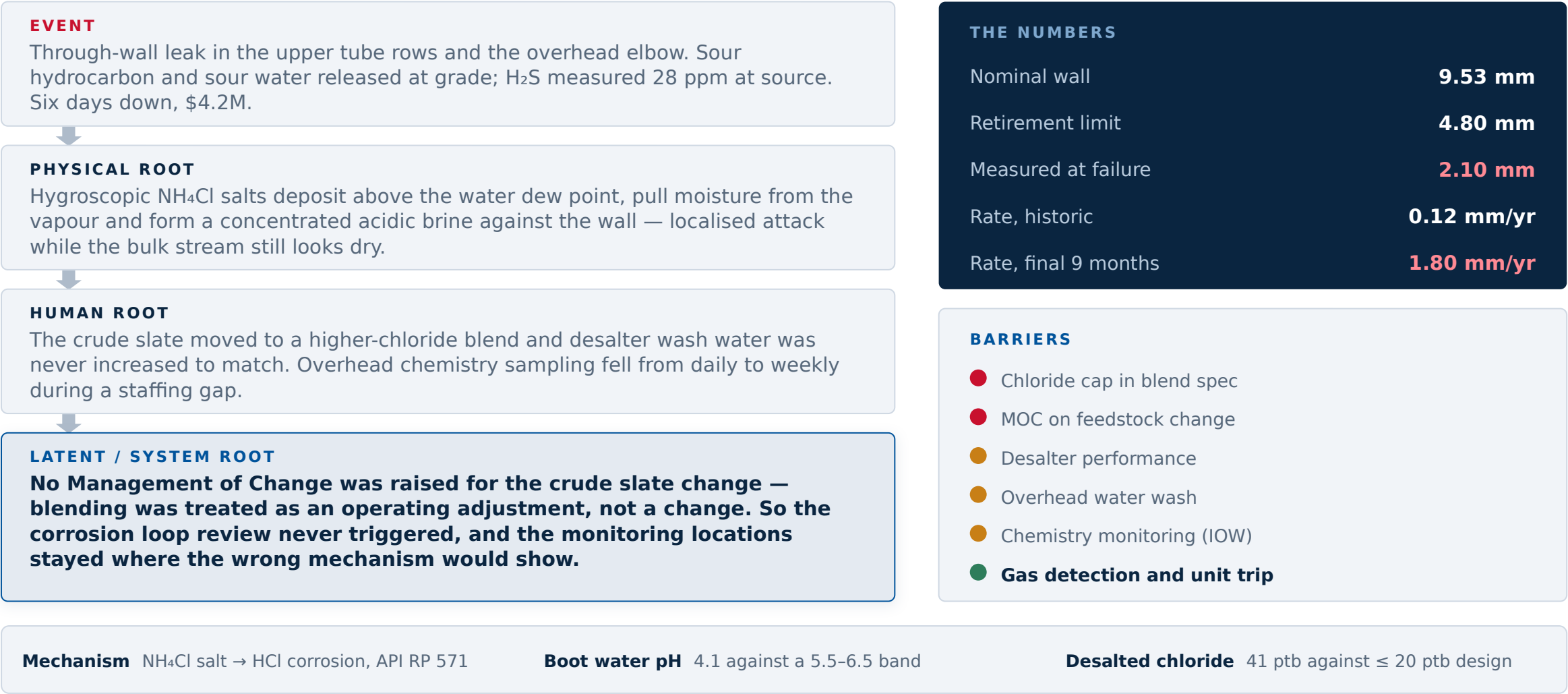
Drawn as a logic tree in the handout; the Five Whys chain runs underneath each one as a narrative caption.

Mechanism naming API RP 571 terminology throughout — never the word “corrosion” on its own

Action ranking Eliminate → substitute → engineering → administrative → PPE

We inspected the wrong part of this line for years.

Ammonium chloride salt corrosion, progressing to hydrochloric acid attack once free water condensed (API RP 571).



This pump told us three times. Nobody asked why.

Mechanical seal failure on 290 °C hydrocarbon service — loss of containment above the flash point, with contractors 30 m away.

EVENT

The seal failed catastrophically; 2.1 m³ of gas oil at 290 °C released and pooled. Gas detection alarmed, the pump tripped, area ESD initiated. It did not ignite. 18 hours down, \$0.9M.

PHYSICAL ROOT

The Plan 11 flush line was blocked with coke fines at the orifice. With no flush the seal chamber lost cooling and pressure margin, the faces ran dry, and the carbon face cracked by thermal shock.

HUMAN ROOT

Three seal failures in eighteen months were each closed as a maintenance job — like-for-like, no investigation. The vibration alarm was acknowledged and reset without a work order.

LATENT / SYSTEM ROOT

No defect elimination process and no bad-actor threshold. Nothing said that three failures in eighteen months is abnormal, so the repeats were never visible as a pattern — only as a series of small, well-handled jobs.

THE NUMBERS

Seal arrangement	API 682 Arr. 1
Vibration at failure	11.4 mm/s
ISO 10816-3 alarm	7.10 mm/s
Failures in 18 months	3
MTBF, achieved / target	6 / 36 mo

BARRIERS

- Seal flush (Plan 11)
- Dual seal containment
- Bad actor / RCA trigger
- Vibration alarm response
- Ignition control — untested
- Gas detection and ESD

Mechanism Seal face thermal shock from loss of flush

Service 290 °C, above the 195 °C flash point

Fix API 682 Arr. 2 dual seal, Plan 53B barrier fluid

The instrument that would have seen this coming was switched off.

Creep and stress rupture from localised overheating with internal coke deposition (API RP 571), on a 9Cr-1Mo radiant tube.

EVENT

A Pass 3 radiant tube bulged and ruptured longitudinally. Crude discharged into the firebox and ignited. Heater tripped, emergency depressuring initiated, refractory and adjacent tubes damaged. 21 days down, \$18.0M.

PHYSICAL ROOT

A partly plugged burner tip impinged flame on the tube. Coke laid down inside; coke insulates, so heat stayed in the wall instead of reaching the process; metal temperature climbed, which laid down coke faster.

HUMAN ROOT

The Pass 3 tube skin thermocouple failed and was inhibited in the DCS to stop nuisance alarms — eight months before the rupture. Pass flow imbalance had been normalised on the panel.

LATENT / SYSTEM ROOT

Instrument inhibits could be applied indefinitely with no register, no expiry and no risk assessment. A reasonable person solved a real problem, and the organisation provided no mechanism that would ever ask for the measurement back.

THE NUMBERS

Tube material	A335 P9
Design TMT	620 °C
Estimated peak TMT	760 °C
Internal coke layer	6 mm
Diameter increase	6.2%

BARRIERS

- Tube metal temperature monitoring
- Integrity Operating Window
- Condition-based decoking
- Pass flow balancing
- IR thermography survey
- Trip and emergency depressuring

Mechanism Creep / stress rupture, API RP 571

Damage indicator > 3% diameter increase; 6.2% measured

Fix Redundant TMT, inhibit register, API RP 584 IOW

WHAT THE THREE HAVE IN COMMON

Every one had a competent engineering answer and an uncomfortable organisational one underneath it.

Not one of these was caused by somebody being careless on the day.

RCA-01 Overhead

PHYSICAL ROOT

NH₄Cl under-deposit corrosion

LATENT ROOT

Crude slate change without MOC

"Inspecting the wrong location"

RCA-02 Pump seal

PHYSICAL ROOT

Dry-running seal faces

LATENT ROOT

No bad-actor threshold

"Fixing without asking why"

RCA-03 Heater tube

PHYSICAL ROOT

Creep from localised overheating

LATENT ROOT

Inhibit with no expiry

"A measurement switched off"

The finding. Each was caused by a system that permitted a reasonable local decision to remove a barrier permanently, and had no route by which anyone would notice. Fixing three pieces of equipment does not fix that. Everything that follows exists to close it.

Hierarchy of controls Every corrective action ranked — eliminate and substitute before administrative

PPE Appears in none of the three action tables. No respirator prevents a condenser from corroding.

02

THE FORWARD LOOK

What could still fail

Three of the five entries are the equipment that just failed. The RCA findings have to land somewhere, on somebody, on a date.

Three of these five are the equipment you just watched fail.

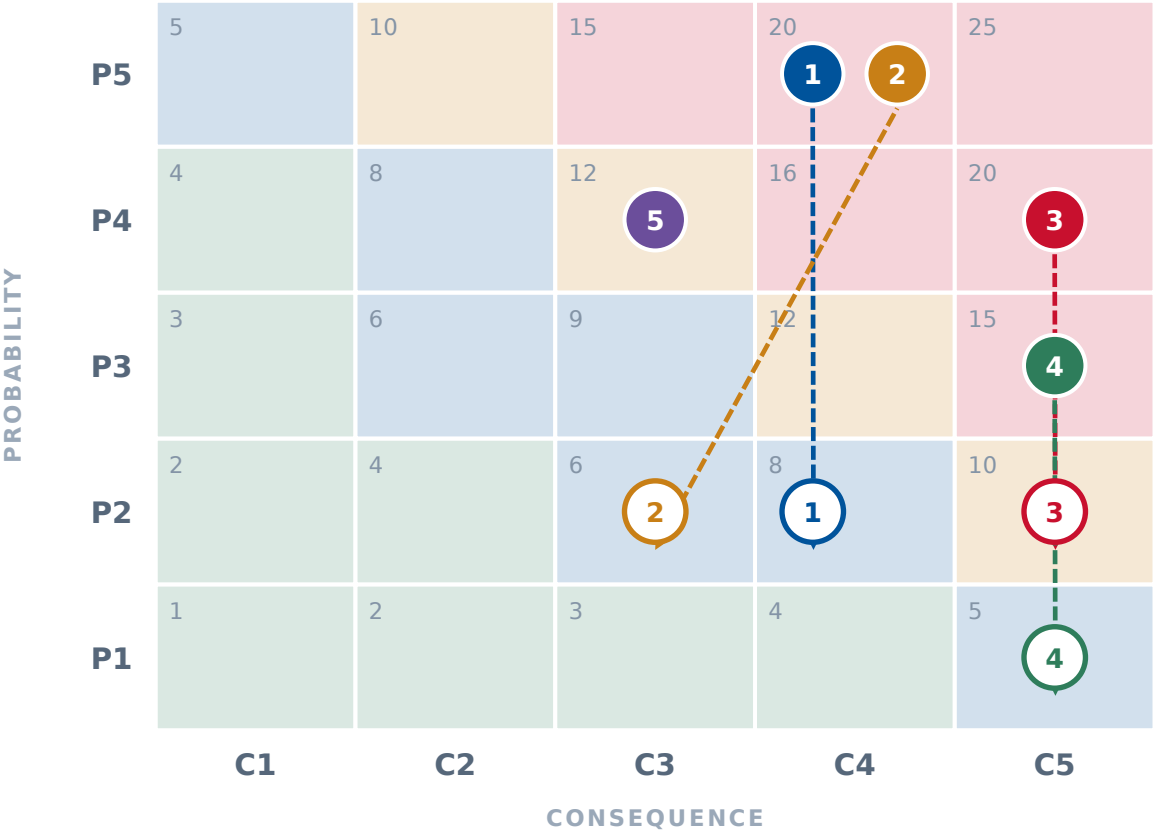
Scored twice — once with no credit for controls, once after them. The gap between the two is what the control spend buys.

RR-01 <i>failed</i>	E-1103A overhead condenser Under-deposit corrosion perforates tubes — sour hydrocarbon and H ₂ S at grade	20 CRITICAL	→	8 MEDIUM	Inspection Engineer review: Quarterly
RR-02 <i>failed</i>	P-1104A HGO pump Seal failure releases 290 °C hydrocarbon above its flash point — pool fire	20 CRITICAL	→	6 MEDIUM	Rotating Equipment Engineer review: Monthly
RR-03 <i>failed</i>	H-1101 radiant tubes Creep rupture discharges crude into the firebox with escalation potential	20 CRITICAL	→	10 HIGH	Fired Heater Specialist review: Monthly
RR-04 <i>by design</i>	T-1101 flash zone shell Sulfidation wall loss — catastrophic loss of tower inventory	15 CRITICAL	→	5 MEDIUM	Inspection Manager review: Annual
RR-05 <i>no cover</i>	E-1108 cooling water side Tube leak admits hydrocarbon to the cooling tower — flammable release	12 HIGH	→	12 HIGH	Utilities Engineer review: Monthly

RR-05 does not move. Every barrier on that line is broken, overdue or not installed, so none of them earns a reduction. A register that quietly scored it down would be lying — and the identical numbers are the strongest argument for funding it.

Four Critical entries become one High and three Medium. One does not move at all.

Solid marker = inherent risk, with no credit for controls. Ringed marker = residual, after the barriers are in place and verified.



- 1 E-1103A overhead condenser**
20 Critical → 8 Medium
- 2 P-1104A HGO pump**
20 Critical → 6 Medium
- 3 H-1101 radiant tubes**
20 Critical → 10 High
- 4 T-1101 flash zone**
15 Critical → 5 Medium
- 5 E-1108 cooling water side**
12 High → 12 High

Marker 3 stays High. Marker 5 does not move.
A fired heater on crude service carries irreducible consequence — the answer is continued active management, not a lower score.

03

AUTHORISATION

Controlling the work the register created

Eleven actions on live hydrocarbon equipment. Every one is more dangerous to perform than to leave alone, for the duration of the job.

A permit to work is not a form. It is who may do what, where, and under what conditions.

A site can have an excellent form and no system. The form is only the evidence that the architecture was applied.

LIFECYCLE



PERMIT TYPES

Hot Work Type A · spark or flame

Hot Work Type B · hot surface, vehicle entry

Cold Work · mechanical, no ignition source

Line Breaking · any breach of an envelope

Confined Space · continuous monitoring

Electrical / LOTO · duration of isolation

Excavation · revalidated daily

Working at Height · above 1.8 m

Radiography · per exposure window

ROLES · SEPARATION OF DUTIES

Performing Authority
scope, JSA, controls in the field

Issuing Authority
verifies isolations and gas tests

Area Authority
plant condition, cross-permit conflicts

Isolating Authority
applies and proves isolations

Authorised Gas Tester
all atmospheric testing

Site Controller
SIMOPS and emergency authority

No one person may hold PA and IA on the same permit.

SUPPORTING CERTIFICATES

Isolation certificate
every point listed, locked, proved dead

LOTO register
a personal lock for every exposed person

Confined space entry
atmosphere, standby, rescue plan

Excavation
services located, shoring, access

Radiography
barriers, dose rates, timing

The permit authorises the work. The certificates prove the conditions that make it safe — each signed by a different competent person.

Analogy, used once: owning a key is not the same as being authorised to use it at a particular moment.

THE THREE THAT MATTER

Every site has a permit form. Incident histories cluster on the same three failures.

A system that does not address these explicitly is a filing exercise.

01

Cross-referencing and SIMOPS

Two permits, each individually safe, become dangerous together.

- A permit map — every live permit plotted on the unit plot plan, on the board and mirrored electronically
- A SIMOPS matrix classifying every activity pair green, amber or red
- A daily coordination meeting before shift start, chaired by the Site Controller
- Cross-reference fields on the permit; the Area Authority signs the deconfliction

02

Shift handover and validity

The night shift inherits an assumption instead of a verified condition.

- No permit is valid beyond the shift that issued it — 12 hours maximum, regardless
- A permit continuing is suspended and revalidated: the oncoming IA walks the job and re-checks every isolation
- Outgoing and oncoming IA review the live permit register face to face at the board
- Any permit suspended for an ESD or gas alarm is cancelled, not revalidated. It starts again

03

The audit and assurance loop

Does what was written on the permit match what is happening on the ground?

- Level 1 — field verification by the Issuing Authority, every permit, every shift
- Level 2 — supervisory spot audit, 10% of live permits, weekly
- Level 3 — independent HSE assurance, monthly, plus annual external
- Leading indicators tracked monthly — and audit findings become risk register entries

Applied on the live permit A hot work permit 22 m away was suspended for the duration of the line break — red on the SIMOPS matrix, not a judgement call on the day

One live permit, for a job that came off the register, on the pump that failed.

Cold Work, cross-referenced to Line Break LB-2026-0119 and Isolation Certificate IC-2026-0286.

PERMIT TO WORK · COLD WORK

No. PTW-CDU-2026-0473

Equipment	P-1104A · Heavy Gas Oil Pump
Register ref	RR-02 interim action, pending the TA-2027 dual seal upgrade
Work	Replace seal cartridge; fit filtered flush and chamber pressure transmitter; align and recommission
Valid	06:30 - 18:30, 02 July 2026 — one shift only, 12 h maximum
Isolations	Suction and discharge DBB locked closed, bleeds to closed drain; flush line blocked; M-1104A racked out, 6 personal locks
Gas test	06:05 · LEL 0% · H ₂ S 0 ppm · O ₂ 20.9% · instrument MX6-4471, cal. due 18 Sep 2026

SIMOPS SCREEN

PTW-CDU-2026-0468 scaffolding, pump row B — GREEN

PTW-CDU-2026-0471 hot work, pipe rack 4 (22 m) — RED, SUSPENDED

Signatures · PA A. O. Joy 06:10 IA I. M. Olamide 06:25 AA D. Christopher 06:28 AGT A. G. Y. Aondoakaa 06:05

Handback accepted 17:35 · permit cancelled 17:40 · work order closed in CMMS with failure and repair coding 17:45

Two details worth pausing on

A hot work permit 22 metres away was suspended because the SIMOPS matrix said so — not because somebody happened to notice on the day.

And the last line

Closing the work order with failure and repair coding sends this job back into the reliability system — exactly the feedback RCA-02 found missing.

04

EXECUTION

At the hands of a fitter with a spanner

The seal replacement on the pump that failed, under the permit just issued. Twelve steps, from isolation to handback.

Twelve steps. Four still carry risk after every control we have.

Same pump as RCA-02, same register entry as RR-02, same permit as PTW-CDU-2026-0473.

1 Verify	2 LOTO	3 Isolate	4 Drain	5 Purge	6 Break	7 Seal out	8 Clean	9 Seal in	10 Align	11 Start	12 Close
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Amber = residual risk still Medium after all controls

3	Verify process isolation <i>Isolating Authority</i>	HAZARD 16" suction holds a standing column of hot gas oil at line pressure; a passing valve looks identical to a closed one	CONTROLS Double block and bleed both sides, locked closed · bleeds cracked to closed drain and watched for 10 minutes with no flow · independent gauge at both ends, not the DCS value
4	Cool down and drain casing <i>PA + Authorised Gas Tester</i>	HAZARD Residual hydrocarbon at 290 °C — 95 °C above its flash point; H ₂ S liberated as it depressures	CONTROLS Casing verified below 60 °C by contact pyrometer before any fitting is loosened · closed drain only · personal H ₂ S monitors alarming at 5 ppm · SCBA staged · approach taken upwind
8	Clean seal chamber and flush line <i>Fitters + HSE Officer</i>	HAZARD Pyrophoric iron sulfide scale — it self-ignites within seconds of drying in air, with no ignition source present	CONTROLS Deposits kept wet from the moment the chamber is opened · never dry-brushed, no compressed air · scrapings straight into a sealed drum of water · extinguisher within reach and manned
11	Pressurise and recommission <i>Operations + PA</i>	HAZARD A new seal fails more often in its first hour than in the next three years, and people are standing next to it	CONTROLS Staged pressurisation with everyone clear of the seal plane — nobody in line with the gland · barrier fluid pressure confirmed before the motor starts · vibration within 4 h against the 7.10 mm/s IOW

HOW TO TELL WHETHER A JSA WAS ACTUALLY WRITTEN

“Wear personal protective equipment” appears nowhere as a control on its own.

An experienced supervisor reads the left column and knows the job was never walked.

WHAT A GENERIC JSA SAYS

WHAT THIS ONE SAYS

4	<i>“Hot surfaces — wear PPE”</i>	→	Verified below 60 °C by contact pyrometer before any fitting is loosened
5	<i>“Purge the equipment”</i>	→	Nitrogen asphyxiation named as a hazard, with O ₂ monitoring alarming below 19.5%
6	<i>“Be careful of dropped objects”</i>	→	Far-side bolts slackened first, so the joint opens away from the fitter
8	<i>“Clean the seal chamber”</i>	→	Pyrophoric scale kept wet, sealed drum, no dry brushing, no compressed air
10	<i>“Align the pump”</i>	→	Pipe strain checked with a dial gauge on the casing, accept below 0.05 mm
11	<i>“Start the pump”</i>	→	Nobody stands in line with the gland during first pressurisation

Why it matters PPE is the last line of defence. A JSA that leans on it has run out of ideas earlier in the hierarchy than it should have.

THE CHAIN, CLOSED

The job that fixes the pump also repairs the system that let it fail three times.

Step 12 codes the work order with failure and repair data — the input to the bad-actor register that RCA-02 found missing.



RCA-02 → RR-02 → PTW-CDU-2026-0473 → JSA-CDU-2026-0473 → CMMS → bad-actor register → RCA

One pump, all the way round. The loop does not end at the repair — it ends by feeding the next analysis.

WHAT WE ARE ASKING FOR

Four decisions, four owners.

Three of these cost almost nothing. The one that costs money is the one that stops an \$18 million failure repeating.

01

Make crude slate change an MOC trigger

With corrosion loop review as a required step. This is the control missing in RCA-01, and it is a change to a procedure, not to the plant.

Procedure only · Two weeks
HSE & Technical Safety Manager

02

Open an instrument inhibit register

Mandatory expiry, risk assessment at application, weekly management review of everything live. Eight months of silence killed a heater tube.

No capital · One month
Instrument Engineer

03

Set a bad-actor threshold

Two failures in twelve months triggers an automatic RCA, reviewed monthly. P-1104A failed three times and the system had no way to add them up.

No capital · One month
Maintenance Manager

04

Fund the three actions on RR-05

Hydrocarbon-in-cooling-water analyser, biocide dosing loop repair, overdue eddy current inspection. It is the one register entry that has not moved.

~\$420k indicative · Q3-Q4 2026
Utilities Engineer

Three failures cost \$23.1 million in twelve months. Three of these four asks are procedural and cost nothing but attention.

Acronyms, standards, and where the detail lives.

Every acronym in this deck was defined on the slide where it first appeared. This is the safety net.

ACRONYMS

AGT Authorised Gas Tester	CDU Crude Distillation Unit
CMMS Maintenance Management System	DBB Double Block and Bleed
ESD Emergency Shutdown	HGO Heavy Gas Oil
IOW Integrity Operating Window	JSA Job Safety Analysis
LEL Lower Explosive Limit	LOC Loss of Containment
LOTO Lock Out Tag Out	MOC Management of Change
MTBF Mean Time Between Failures	PTW Permit to Work
RBI Risk-Based Inspection	RCA Root Cause Analysis
SIMOPS Simultaneous Operations	TML Thickness Monitoring Location
TMT Tube Metal Temperature	ptb Pounds per thousand barrels

STANDARDS REFERENCED

- API RP 571**
Damage mechanisms in refining
- API RP 580 / 581**
Risk-based inspection
- API RP 584**
Integrity operating windows
- API RP 573**
Inspection of fired heaters
- API 682**
Pump shaft sealing systems
- API 579-1 / ASME FFS-1**
Fitness-for-service
- ISO 10816-3**
Mechanical vibration evaluation
- ISO 45001**
OH&S management systems

The handout carries the full artefacts: three complete RCAs with barrier analysis and ranked corrective actions · the full risk register with mechanisms, barriers, IOWs and action dates · the PTW system description and live permit · all twelve JSA steps

Group C (29 Members) · Thank you — questions